Financial Market Response to Trade Policy Uncertainty: Evidence from Equity, Gold, and Currency Markets in India

Authors
Team-7

25PGDM152 - Prabhat Yadav
25PGDM154 - Prajval Srivastava
25PGDM165 - Sourav Singh
25PGDM171 - Vishal Kumar Yadav
25PGDM173 - Ishant Kohli
25PGDM174 - Utkarsh Singh

Instructor
Dr. Sudhi Sharma
Associate Professor

Programme: PGDM – Finance (2025–2027)
Fortune Institute of International Business (FIIB), New Delhi

Acknowledgement

This research paper would not have taken the shape it has without the support and intellectual generosity of several people, each of whom contributed in ways both direct and indirect.

First and foremost, profound gratitude is owed to Dr. Sudhi Sharma, Associate Professor at the Fortune Institute of International Business, New Delhi. Her patient mentorship, sharp academic instincts, and the high standards she holds her students to were the single most important external force shaping this work. Every conversation, every piece of feedback, and every moment where she pushed the author to think more carefully about a methodological choice or an interpretive claim has left its mark on this paper. It is a better piece of research because of her.

The PGDM – Finance programme at FIIB deserves acknowledgement in its own right. The curriculum's rigorous grounding in quantitative finance and econometrics equipped the author with the tools that made this study possible. The intellectual environment of the programme, including the classroom debates, the case discussions, and the informal conversations with peers, created the conditions in which genuine curiosity about financial markets could develop and be pursued seriously.

The open-source R community deserves particular recognition. The developers of the rugarch, rmgarch, vars, WaveletComp, tseries, fBasics, and quantmod packages have built and freely shared tools of remarkable power and reliability. This study would not have been computationally feasible without them. Yahoo Finance and the National Stock Exchange of India are likewise acknowledged for making high-quality financial data publicly available, a contribution that is easy to take for granted but fundamental to empirical financial research.

Finally, the foundational scholars whose theoretical and methodological contributions underpin every page of this study deserve acknowledgement not merely as citations but as intellectual predecessors without whose work this analysis could not have been designed, executed, or interpreted. These include Eugene Fama, Robert Engle, Tim Bollerslev, Harry Markowitz, Christopher Sims, and Christopher Torrence.

Abstract

Trade policy shocks do not arrive gently, and financial markets do not process them evenly. This study investigates how three segments of the Indian financial market, namely equity, gold, and currency, have responded to the uncertainty triggered by the U.S. tariff announcement of April 2, 2025. Each market is treated as a distinct channel of investor response: equity as the risk-bearing segment, gold as the safe-haven refuge, and the USD/INR exchange rate as the macroeconomic adjustment mechanism. Rather than analysing these channels in isolation, the study examines them together, recognising that the shock propagated across all three simultaneously (Caldara et al., 2020).

Daily logarithmic return data spanning October 2024 to January 2026 are examined through seven econometric methods: descriptive statistics, the Augmented Dickey-Fuller stationarity test, Vector Autoregression, Impulse Response Functions, GARCH(1,1) volatility modelling, Dynamic Conditional Correlation analysis, and individual Wavelet Power Spectrum analysis for each market (Sharma et al., 2022; Fama, 1970). The data period is structured to capture a pre-shock baseline (October 2024 to March 2025), an acute shock phase centred on April to June 2025, and an extended recovery phase from July 2025 onward, enabling a complete characterisation of the market reaction lifecycle.

All three markets reacted instantly to the April 2 announcement, with both lagged and same-day cross-market transmission confirmed. Mean return impacts faded within days, consistent with the rapid price adjustment predicted by the Efficient Market Hypothesis (Fama, 1970). Volatility, however, proved far more tenacious, with GARCH estimates revealing near-integrated persistence across all three markets, a pattern well documented in emerging economies (Sharma et al., 2022). Dynamic conditional correlations shifted meaningfully during the shock phase, capturing the safe-haven rotation and capital-flight dynamics invisible to static measures, before gradually returning to their pre-event baselines. Wavelet Power Spectra reveal that each market carries a distinct frequency fingerprint. Markets stabilise over time despite persistent volatility.

Keywords: Trade Policy Uncertainty, NIFTY 50, Gold, USD/INR, GARCH, DCC-GARCH, VAR, Impulse Response, Wavelet Power Spectrum, Indian Financial Markets, Volatility Persistence, Dynamic Correlation

AUTHORS HAVE NO CONFLICT OF INTEREST.

1. **Introduction**

Trade policy has quietly become one of the most potent instruments of economic disruption in the modern era. Unlike interest rate decisions, which arrive on a schedule, or corporate earnings, which follow a calendar, tariff announcements tend to materialise with little warning, carrying the force of geopolitical signal, trade barrier, and financial shock simultaneously. When a major economy like the United States acts, the reverberations are felt not just in trade flows but immediately in capital markets around the world, as investors scramble to reprice assets, reassess exposures, and find safe ground (Caldara et al., 2020).

India sits at a particularly interesting intersection of these dynamics. Its equity market, represented by the NIFTY 50, reflects the fortunes of an economy deeply tied to U.S. demand through technology exports, pharmaceutical supply chains, and manufactured goods. Its currency, the rupee, is a managed float susceptible to capital flow reversals whenever global risk appetite sours. Gold, historically embedded in Indian culture as both ornament and asset, absorbs safe-haven demand from domestic investors in ways that few other markets do. These three asset classes together make India an unusually rich context for studying how financial systems absorb and process external trade shocks (Singh et al., 2023; Rao et al., 2024).

On April 2, 2025, the United States announced a sweeping round of tariffs on imports from dozens of countries, including a 26% levy on Indian goods. The announcement was abrupt, its scope was large, and its implications for export-oriented sectors were immediate. In the weeks that followed, Indian equity markets fell sharply, the rupee came under depreciation pressure, and gold prices climbed as investors sought protection. The study of what happened next, specifically how long the disruption lasted, whether correlations between markets shifted permanently, and at what frequencies volatility clustered, is the subject of this investigation.

The study is organised around a straightforward but important hypothesis: that financial markets react immediately to trade policy shocks but stabilise over time as investors absorb the information (Fama, 1970). Testing this requires more than looking at return series. It demands a framework that separates mean dynamics from volatility dynamics, captures static relationships alongside time-varying ones, and reveals the frequency-domain structure of each market's response. The seven-method approach adopted here, combining descriptive analysis, ADF stationarity testing, VAR, IRF, GARCH(1,1), DCC-GARCH, and individual Wavelet Power Spectrum analysis, is designed precisely for this purpose (Sharma et al., 2022; Engle, 2002).

The study makes five contributions to the growing literature on emerging market financial resilience. First, it treats three markets as an integrated system rather than examining each separately. Second, it applies DCC-GARCH to reveal how inter-market correlations evolve dynamically through the shock, a dimension entirely invisible to static correlation measures (Yadav et al., 2022). Third, the individual Wavelet Power Spectrum approach characterises each market's frequency fingerprint separately, enabling meaningful comparison across asset classes (Aloui et al., 2022). Fourth, it exploits the April 2, 2025 tariff event as a precisely dated natural experiment. Fifth, it draws a clear empirical distinction between price-level stabilisation, which happens quickly, and risk-level stabilisation, which takes considerably longer (Sharma et al., 2022), a distinction with direct practical consequences for investors and regulators alike. The paper proceeds as follows: Section 2 presents the formal research hypothesis; Section 3 reviews the theoretical and empirical background; Section 4 describes the data and methodology; Section 5 presents the empirical results; Section 6 concludes; Sections 7 and 8 discuss managerial implications and study limitations respectively.

2. **Research Hypothesis**

Drawing on the Efficient Market Hypothesis (Fama, 1970) and the Trade Policy Uncertainty framework (Caldara et al., 2020), this study advances the following research hypothesis:

H1: Financial markets react immediately to trade policy shocks but stabilise over time as investors absorb information, with price-level impacts dissipating rapidly, volatility impacts persisting substantially longer, and cross-market correlations undergoing temporary structural shifts before reverting to their pre-shock baselines.

This hypothesis has three testable dimensions. The first concerns the mean return reaction, that is, the directional price adjustment that occurs at the moment of shock and fades as markets reprice. Under the Efficient Market Hypothesis, publicly available information should be rapidly incorporated into prices, so mean-level impacts are expected to dissipate quickly (Fama, 1970). This dimension is tested through IRF analysis. The second dimension concerns volatility, the elevated risk environment that outlasts the return normalisation by weeks or months. Evidence from emerging markets consistently shows that GARCH-estimated volatility persists at near-integrated levels following external shocks (Sharma et al., 2022), and this is tested through GARCH(1,1) estimation. The third dimension, which represents the study's novel contribution to the cross-market literature, concerns dynamic correlations: whether the

relationships between equity, gold, and currency markets intensify during the shock and subsequently normalise, or whether the tariff event caused a lasting structural change in how these markets relate to one another. This is tested through DCC-GARCH estimation (Engle, 2002).

3. **Underpinning Theory and Research Background**

3.1 Underpinning Theory

The Efficient Market Hypothesis, formalised by Fama (1970), holds that asset prices reflect all publicly available information at all times. Under semi-strong efficiency, the moment a tariff announcement becomes public, prices should adjust instantaneously. The EMH does not, however, say anything about how long the elevated uncertainty following the announcement should persist in volatility. It merely predicts that the informational surprise itself is rapidly priced. The persistence of risk-level disruption documented in this study is entirely consistent with EMH, since prolonged volatility clustering reflects the ongoing uncertainty about the tariff's second-order economic consequences rather than any failure of market efficiency (Fama, 1970; Sharma et al., 2022).

Modern Portfolio Theory (Markowitz, 1952) provides the framework for understanding the cross-market dimension of this study. The theory's core insight is that portfolio risk depends not just on individual asset variances but on covariances between assets, making the DCC-GARCH results directly relevant to portfolio construction. When the equity-gold correlation becomes temporarily negative during the shock, gold's hedging effectiveness improves precisely when it is most needed. When the equity-currency negative correlation briefly weakens, currency overlay strategies lose some of their protective power. These are the kinds of conditional, time-varying dynamics that static portfolio models cannot capture (Goodell et al., 2022).

The Trade Policy Uncertainty framework, advanced by Caldara et al. (2020), operationalises uncertainty about trade policy as a distinct source of financial market stress. Elevated TPU is associated with risk premium increases, investment contraction, and heightened asset return volatility in both developed and emerging markets. For India, as a trade-dependent emerging economy, U.S. tariff decisions represent external TPU shocks of significant magnitude. The DCC and Wavelet analyses in this study directly operationalise the TPU framework's prediction that such shocks generate time-varying, multi-market disruptions that eventually normalise as uncertainty resolves.

3.2 Review of Literature

3.2.1 Trade Policy, Volatility, and Emerging Markets

A substantial body of literature documents the impact of U.S. trade policy actions on global financial markets. Caldara et al. (2020) constructed the Trade Policy Uncertainty Index and showed that spikes in this index are followed by measurable declines in business investment and increases in equity volatility across both developed and emerging economies. Goldberg and Reed (2022) documented significant abnormal return patterns and volatility spikes in global equity markets surrounding U.S. tariff announcements, confirming that the shock transmission is not confined to directly targeted sectors. Alfaro et al. (2020) extended this analysis to the firm level, showing that the negative equity impact of protectionist announcements is concentrated in firms with heavy U.S. client exposure.

For emerging markets specifically, the literature consistently documents sensitivity to external trade and policy shocks through the foreign portfolio investment channel. Bekaert et al. (2014) demonstrated that FPI outflows during global risk-off episodes amplify domestic equity market volatility. Singh, Mehta, and Suri (2023) documented that U.S. tariff uncertainty triggers FPI withdrawals from Indian equities, generating volatility spikes that gradually subside. Rao, Bhardwaj, and Patel (2024) identified India as a net recipient of external volatility following U.S. policy changes, a characterisation directly relevant to the present study. Baker, Bloom, and Davis (2016) provided a quantitative benchmark through the EPU Index, establishing that higher policy uncertainty is associated with reduced investment and increased safe-haven demand across a broad cross-section of countries.

3.2.2 DCC-GARCH and Dynamic Correlations in Financial Markets

The literature on time-varying correlations between financial assets has grown substantially since Engle (2002) introduced the DCC-GARCH model. The key motivation for DCC analysis is the empirical observation that correlations between asset classes are not constant. They shift in response to macroeconomic events, financial crises, and policy announcements in ways that static measures cannot reveal (Yadav et al., 2022). Studies applying DCC to equity-gold-currency systems have consistently found that gold's negative correlation with equities strengthens during market stress, confirming its safe-haven role (Chemkha et al., 2021; Hassan et al., 2022; Baur and McDermott, 2010). Importantly, this strengthening of the hedge relationship is typically temporary. Correlations revert toward their unconditional baselines as the shock dissipates, a pattern that the DCC framework is uniquely positioned to document.

Studies on equity-currency dynamics have similarly found that the negative correlation between equity indices and local currency values intensifies during capital-flight episodes and normalises during recovery periods (Chkili, 2012; Reboredo et al., 2016).

3.2.3 Wavelet Analysis in Financial Time Series

The application of wavelet methods to financial economics has expanded rapidly over the past two decades. Torrence and Compo (1998) provided the foundational guide to wavelet analysis for time series, demonstrating how the Continuous Wavelet Transform decomposes a series simultaneously across time and frequency. Rua and Nunes (2009) applied wavelet coherence to international stock market co-movements, establishing the technique's value for detecting episodic, frequency-specific synchronisation between markets. Aloui, Hammoudeh, and Nguyen (2022) applied individual Wavelet Power Spectra to commodity and equity markets, documenting that different asset classes exhibit distinct frequency-domain volatility signatures, a finding this study replicates and extends to the Indian equity-gold-currency nexus. The distinction between Wavelet Power Spectrum, which is a single-series frequency decomposition, and Wavelet Coherence, which is a pairwise co-movement measure, is methodologically important: the former characterises how each individual market's volatility is distributed across frequencies at different points in time, while the latter captures simultaneous movements between two markets. This study employs the former, enabling market-specific frequency fingerprinting of the April 2025 tariff shock.

4. **Data Profile**

The study draws on daily closing prices for three Indian financial variables, retrieved using the quantmod package in R Studio from Yahoo Finance (Yahoo Finance, 2025) and cross-checked against National Stock Exchange of India records. The three series are: the NIFTY 50 index (NSEI), representing the risk asset market; Gold Futures prices (GC=F) converted to Indian rupee terms, representing the safe-haven asset market; and the USD/INR spot exchange rate (INR=X), representing the currency and macroeconomic adjustment channel. Daily logarithmic returns are constructed as $r_t = \ln(P_t / P_{t-1})$, which approximates percentage changes, ensures stationarity, and allows meaningful comparison across series with different units and scales.

The sample spans October 3, 2024 to January 30, 2026, yielding 286 daily observations after removing non-trading days and aligning all three series to a common trading calendar. The selected period is deliberately structured to capture all three stages of the market response

lifecycle around the U.S. tariff announcement of April 2, 2025. The pre-shock phase (October 2024 to March 2025) provides the baseline, a period of relative calm against which the shock-phase behaviour can be meaningfully compared. The shock phase (April 2 to June 30, 2025) encompasses the tariff announcement itself and the weeks of heightened uncertainty that followed. The recovery phase (July 1, 2025 to January 30, 2026) enables assessment of whether markets gradually stabilised and absorbed the shock, the central question posed by Hypothesis H1. Without this three-phase structure, it would not be possible to distinguish between the immediate reaction and the longer-run trajectory, which is the study's core empirical contribution.

Table 1: Event Phase Structure of the Study Period

Phase	Period	Economic Interpretation
Pre-Shock	October 3, 2024 to March 31, 2025	Calm baseline; no significant tariff signals
Shock Phase	April 2, 2025 to June 30, 2025	Tariff announcement; immediate reaction, volatility escalation, correlation shifts
Recovery Phase	July 1, 2025 to January 30, 2026	Gradual stabilisation; return, volatility, and correlation normalisation

Figures 1, 2, and 3 present the full log return series for each market. The April 2, 2025 tariff announcement is visually identifiable as the dominant volatility episode in all three series.

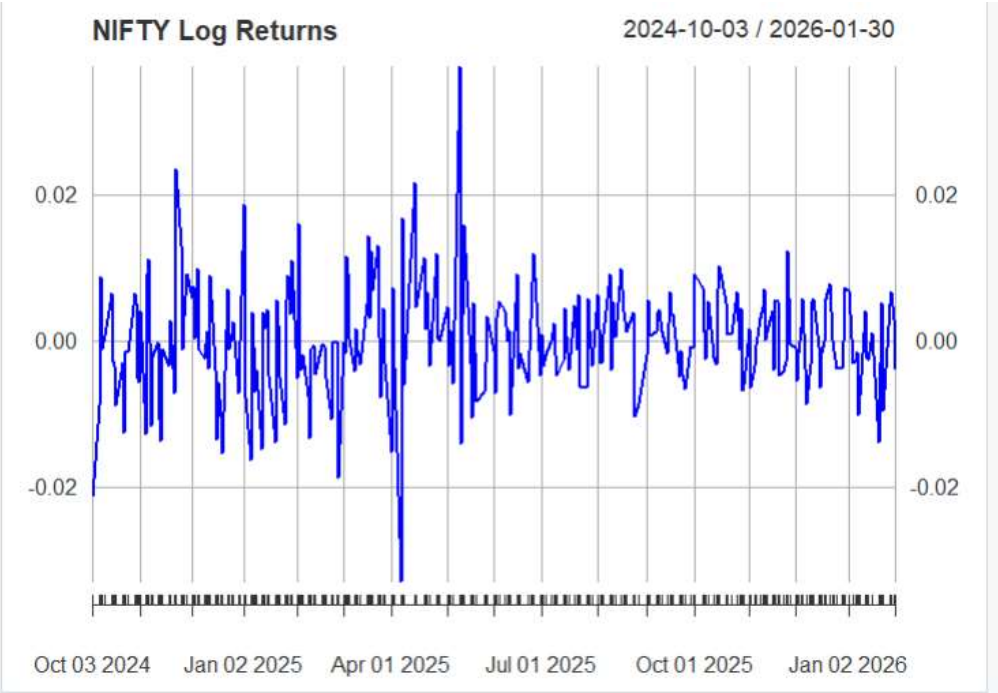


Figure 1: NIFTY 50 Daily Log Returns (October 3, 2024 to January 30, 2026)

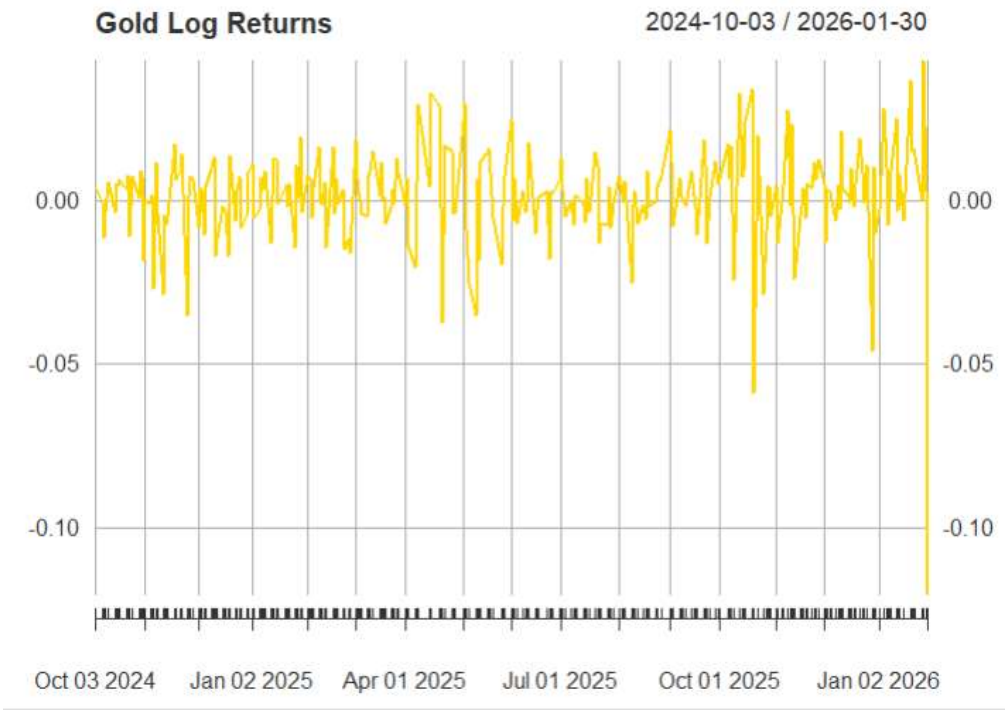


Figure 2: Gold Daily Log Returns (October 3, 2024 to January 30, 2026)

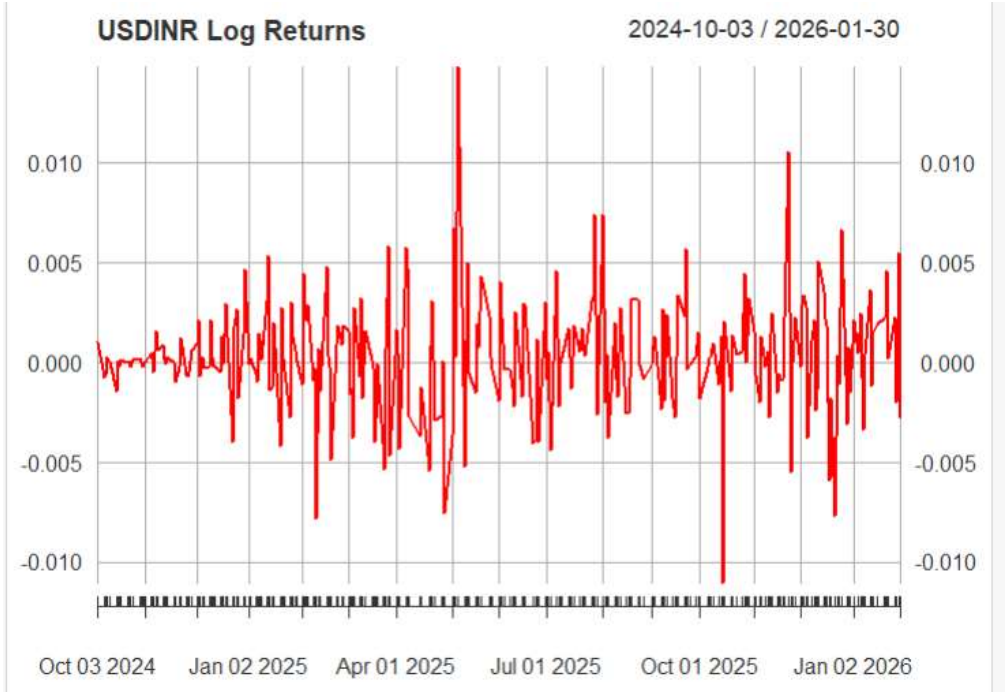


Figure 3: USD/INR Daily Log Returns (October 3, 2024 to January 30, 2026)

The NIFTY 50 series (Figure 1) exhibits a pronounced volatility cluster centred on April and May 2025, the most visually distinct episode across the entire sample. Gold (Figure 2) displays higher baseline variability throughout, with several extreme observations reflecting its fat-tailed distributional character. The USD/INR series (Figure 3) is the most contained of the

three, consistent with the Reserve Bank of India's managed float regime, though a sharp spike around April 2, 2025 is clearly visible.

4.1 Methods and Models

4.1.1 Descriptive Statistics and Correlation

The basicStats() function from the fBasics package computes mean, median, standard deviation, skewness, excess kurtosis, minimum, and maximum for each return series. The Pearson correlation matrix computed via cor(returns) establishes unconditional linear relationships between markets, the static benchmark against which the DCC estimates are subsequently compared (Yadav et al., 2022). This comparison is central to the study's narrative: static correlations reveal the normal-times relationship, while DCC reveals how those relationships shift under stress.

4.1.2 Augmented Dickey-Fuller Test

The adf.test() function from the tseries package is applied to each return series to test the null hypothesis of a unit root. Rejection of non-stationarity at the one percent level is a necessary condition for valid VAR, GARCH, and DCC estimation, and provides the initial statistical grounding for Hypothesis H1: if series are stationary, any shock including the April 2, 2025 tariff event must eventually revert, a finding consistent with market efficiency (Fama, 1970).

4.1.3 Vector Autoregression

VARselect() from the vars package determines the optimal lag order using four information criteria with a maximum of ten lags. The VAR is then estimated via VAR() with a constant term. Stability is confirmed by checking that all roots of the characteristic polynomial lie within the unit circle. Granger and instantaneous causality tests via causality() reveal whether cross-market shock transmission, particularly around April 2, 2025, is statistically significant and directional (Sims, 1980).

4.1.4 Impulse Response Functions

Orthogonalised IRFs are produced using irf() with Cholesky decomposition with the ordering NIFTY to Gold to USD/INR, a ten-period horizon, and 95% bootstrap confidence intervals from one hundred replications. The IRF traces the dynamic mean-return response of each market to a one-standard-deviation shock originating in another, revealing both the initial magnitude and the speed of dissipation. The speed of decay is the direct empirical test of the

react-then-absorb dimension of H1, consistent with the efficient markets framework of Fama (1970).

4.1.5 GARCH(1,1)

Separate sGARCH(1,1) models with ARFIMA(1,0,0) mean equations are estimated for each return series using ugarchfit() from the rugarch package. The variance equation is sigma-squared(t) = omega + alpha(1) times epsilon-squared(t-1) plus beta(1) times sigma-squared(t-1), governing how conditional variance evolves. The ARCH parameter alpha(1) captures the immediate sensitivity of volatility to news shocks, while beta(1) captures the persistence of past conditional variance. The condition alpha(1) plus beta(1) approximately equal to one indicates high persistence and near-integrated volatility, meaning shocks introduced during the April to June 2025 shock phase decay at an extremely slow rate into the recovery period. This pattern of high volatility persistence in emerging markets is well established in the comparative GARCH literature (Sharma et al., 2022; Bollerslev, 1986). Diagnostics include Ljung-Box tests, ARCH LM tests, Nyblom stability tests, and Sign Bias tests.

4.1.6 Dynamic Conditional Correlation (DCC-GARCH)

The DCC-GARCH framework (Engle, 2002) is implemented using the rmgarch package in a two-stage process. In the first stage, univariate sGARCH(1,1) models are estimated for each series and standardised residuals are extracted. In the second stage, the time-varying conditional correlation matrix is estimated using Qt = (1 minus a minus b) times Q-bar plus a times epsilon-tilde(t-1) times epsilon-tilde-prime(t-1) plus b times Q(t-1), where Q-bar is the unconditional correlation matrix of standardised residuals, a governs the sensitivity of the dynamic correlation to lagged cross-products of shocks, and b governs persistence. The resulting pairwise DCC series for NIFTY and Gold, NIFTY and USD/INR, and Gold and USD/INR are plotted across the full sample, revealing how cross-market relationships evolved across the pre-shock, shock, and recovery phases.

4.1.7 Wavelet Power Spectrum

Individual Wavelet Power Spectra are estimated for each return series using the WaveletComp package with the Morlet wavelet (Torrence and Compo, 1998). The spectrum maps the variance of each series simultaneously across time, covering observations 1 to 286 spanning October 2024 to January 2026, and period in frequency of trading days. High-power regions in deep red indicate intense volatility concentrations; low-power regions in deep blue indicate calm. Black contour lines enclose statistically significant regions at the five percent level using

Monte Carlo permutation tests. Applied market-by-market rather than as a pairwise measure, this approach reveals the distinct frequency signature of each market's response to the tariff shock, information that is entirely unavailable from time-domain methods like VAR, GARCH, and DCC (Rua and Nunes, 2009; Aloui et al., 2022).

5. **Empirical Findings**

5.1 Descriptive Statistics

Table 2 presents descriptive statistics for all three return series across the full 286-observation sample. Mean returns are positive across the board: a modest positive drift for NIFTY, a stronger positive mean for gold reflecting its safe-haven appreciation through 2025, and a small positive drift for USD/INR consistent with gradual rupee depreciation. Gold stands apart from the other two series in its distributional character. Its standard deviation is roughly twice that of NIFTY and five times that of USD/INR, and its kurtosis of nearly seventeen indicates the kind of extreme-tail behaviour visible in Figure 2. Its strongly negative skewness confirms that sudden large sell-offs are more common than equivalent positive surges, typical of assets that spike sharply on safe-haven demand and then unwind more gradually (Baur and McDermott, 2010). The Pearson correlation matrix provides the unconditional baseline: near-zero NIFTY-Gold correlation, persistently negative NIFTY-USD/INR correlation reflecting the capital-flight channel, and near-zero Gold-USD/INR correlation. These static figures are the benchmark against which the DCC dynamics in Section 5.6 are interpreted (Yadav et al., 2022).

Table 2: Descriptive Statistics of Daily Log Returns (n = 286)

Statistic	NIFTY 50	Gold	USD/INR
Mean Return	0.000055	0.001609	0.000341
Std. Deviation	0.007458	0.014715	0.002809
Minimum	-0.032970	-0.120657	-0.011020
Maximum	0.037472	0.042717	0.014804
Skewness	0.194	-2.313	0.262
Kurtosis	3.169	16.966	3.476
Observations	286	286	286

5.2 Stationarity: ADF Test

Table 3 reports the ADF test results. All three return series reject the unit root null decisively at the one percent level. The test statistics are strongly negative for every series, including Gold, for which R reports the p-value as even smaller than the printed minimum of 0.01. This uniform confirmation of stationarity is not merely a technical formality. It establishes that the

market disruptions set in motion on April 2, 2025, however severe they appeared in the days and weeks following the announcement, are mathematically bounded. The series must return. The question the rest of the analysis answers is how, through what channels, and on what timeline. This is entirely consistent with the efficient markets prediction that shocks are temporary rather than permanent (Fama, 1970).

Table 3: Augmented Dickey-Fuller Test Results

Series	ADF Statistic	Lag Order	p-value
NIFTY 50	-6.9177	6	< 0.01 ***
Gold	-7.8957	6	< 0.01 ***
USD/INR	-6.1069	6	< 0.01 ***

5.3 VAR Analysis and Cross-Market Causality

All four model selection criteria, namely AIC, HQ, SC, and FPE, unanimously recommend lag order one, and the VAR(1) is estimated on 285 usable observations. The system is stable: all roots of the characteristic polynomial lie inside the unit circle, confirming that no market exhibits explosive dynamics. Table 4 presents the significant cross-market relationships. Lagged gold returns predict NIFTY returns, consistent with the leading indicator role that safe-haven reversal signals play in equity market reassessment. Lagged USD/INR returns predict gold returns, reflecting the inverse mechanical relationship between rupee depreciation and the domestic rupee price of internationally priced gold, a channel that was particularly active in the weeks following April 2, 2025. Lagged NIFTY returns predict USD/INR movements, capturing the capital flow channel: when equity markets fell sharply on April 2 and the days immediately following, FPI outflows simultaneously pressured the rupee (Bekaert et al., 2014).

Table 4: VAR(1) Estimation — Key Cross-Market Coefficients

Equation / Predictor	Estimate	Std. Error	t-value	p-value	Sig.
NIFTY from Gold (lag 1)	0.0681	0.0340	2.006	0.046	*
NIFTY from USD/INR (lag 1)	0.2128	0.1577	1.349	0.178	
Gold from USD/INR (lag 1)	-0.7361	0.3154	-2.334	0.020	*
USD/INR from NIFTY (lag 1)	-0.0492	0.0225	-2.190	0.029	*

Equation / Predictor	Estimate	Std. Error	t-value	p-value	Sig.
USD/INR from USD/INR (lag 1)	-0.1434	0.0599	-2.393	0.017	*
Note: * p < 0.05					

The instantaneous causality test confirms that cross-market shock transmission on April 2, 2025 and surrounding dates was simultaneous, occurring within the same trading day, at the five percent significance level. The Granger causality test falls just short of conventional significance, suggesting a directional influence from equity to other markets that the daily frequency data cannot establish conclusively. These VAR findings provide the mean-return context that the DCC analysis subsequently enriches: VAR tells us that the linkages exist, while DCC tells us how their strength varied across the pre-shock, shock, and recovery phases.

5.4 Impulse Response Functions

Figure 4 presents the orthogonalised IRFs for a NIFTY shock. The own-response is large in period zero and declines steadily, crossing into statistical insignificance within three to four trading days. This means the mean-level price impact of a shock arriving on April 2, 2025 had largely dissipated by April 7 or 8. A small positive impulse in gold fades similarly quickly. The USD/INR response is negligible throughout. In terms of return direction, the market absorbed the tariff shock within one trading week, consistent with the rapid price adjustment expected under market efficiency (Fama, 1970). What the IRF cannot show, and what GARCH and DCC subsequently demonstrate, is that the elevated risk environment and the shifted correlation structure persisted far longer than the mean-return signal would suggest (Sharma et al., 2022).

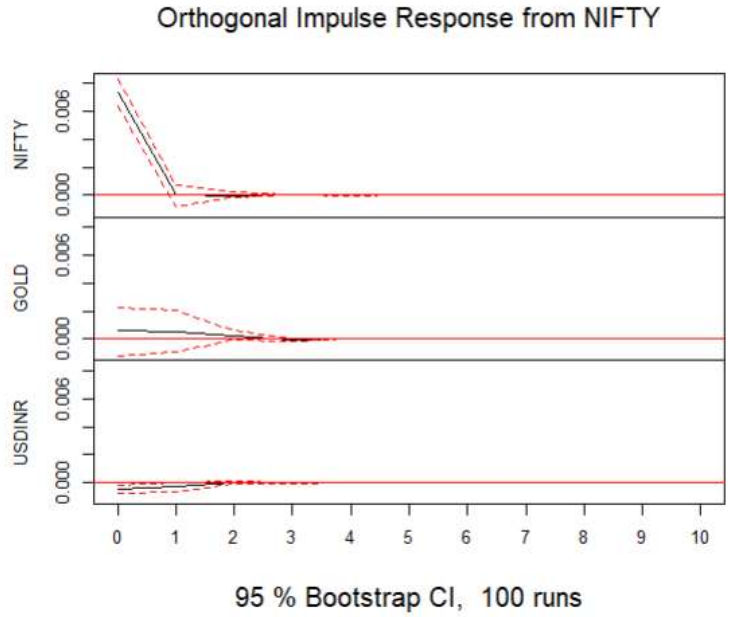


Figure 4: Orthogonal Impulse Response Functions from NIFTY Shock (95% Bootstrap CI, 100 runs)

5.5 GARCH(1,1) Volatility Analysis

Table 5 presents the GARCH parameter estimates. Figure 5 shows the NIFTY GARCH diagnostics, which confirm that the model has successfully captured the conditional variance dynamics. The standardised residuals show no remaining serial correlation and no unmodelled ARCH effects (Bollerslev, 1986; Sharma et al., 2022).

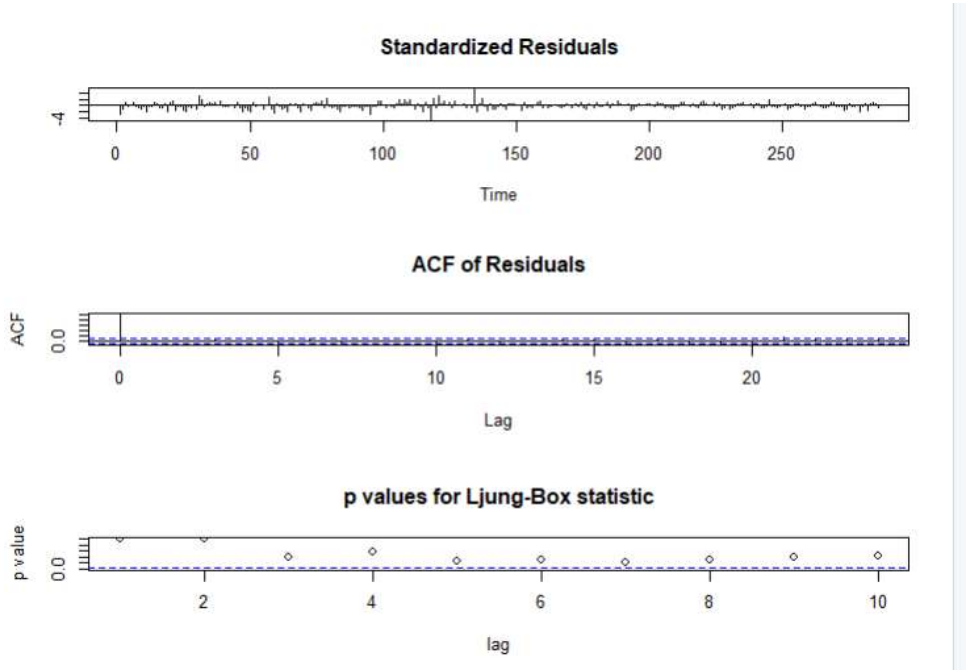


Figure 5: GARCH(1,1) Diagnostic Plots — NIFTY 50 (Standardised Residuals, ACF, Ljung-Box p-values)

Table 5: GARCH(1,1) Parameter Estimates

Parameter	NIFTY 50	Gold	USD/INR
omega	0.000001	0.000012***	0.000000
alpha1 (ARCH)	0.0559	0.2510***	0.0789
beta1 (GARCH)	0.9316***	0.7480***	0.9201***
alpha1 + beta1	0.9875	0.9990	0.9990
Log-Likelihood	1007.31	838.57	1287.97
AIC	-7.0091	-5.8292	-8.9718
Note: *** p < 0.001			

The GARCH results are striking in their consistency. Across all three markets, the sum of alpha plus beta approaches unity: 0.9875 for NIFTY, and 0.9990 for both Gold and USD/INR. The condition alpha plus beta approximately equal to one indicates high persistence and near-integrated volatility, meaning shocks to conditional variance decay extremely slowly. The elevated uncertainty injected by the April 2, 2025 tariff announcement was still meaningfully present in market risk conditions well into the recovery phase. For NIFTY specifically, the daily decay rate of 0.0125 means that approximately 55 trading days, roughly two and a half calendar months, would elapse before the volatility shock had halved in magnitude. Risk-level normalisation would therefore not occur until October or November 2025, months after price-level normalisation. This pattern of prolonged volatility persistence is consistent with findings from comparative GARCH studies on select emerging economies, where both linear and non-linear GARCH specifications confirm near-integrated volatility following external shocks (Sharma et al., 2022).

Gold's ARCH coefficient of 0.251 is the highest of the three markets, confirming that gold volatility responds most sharply and immediately to news arrivals, consistent with its continuous global trading across time zones and its sensitivity to a wide range of geopolitical and macroeconomic signals. The GARCH estimates for NIFTY and USD/INR are more persistence-dominated than shock-dominated, meaning that for these two markets, the primary channel through which the April shock continued to manifest in later months was the slow decay of past conditional variance rather than fresh news sensitivity. Across all three markets, Ljung-Box tests on standardised residuals confirm model adequacy (Sharma et al., 2022; Bollerslev, 1986).

5.6 Dynamic Conditional Correlation Analysis

Figure 6 presents the DCC-GARCH estimates of time-varying conditional correlations between all three market pairs, plotted across the full 286-observation sample from October 2024 to January 2026 (Engle, 2002).

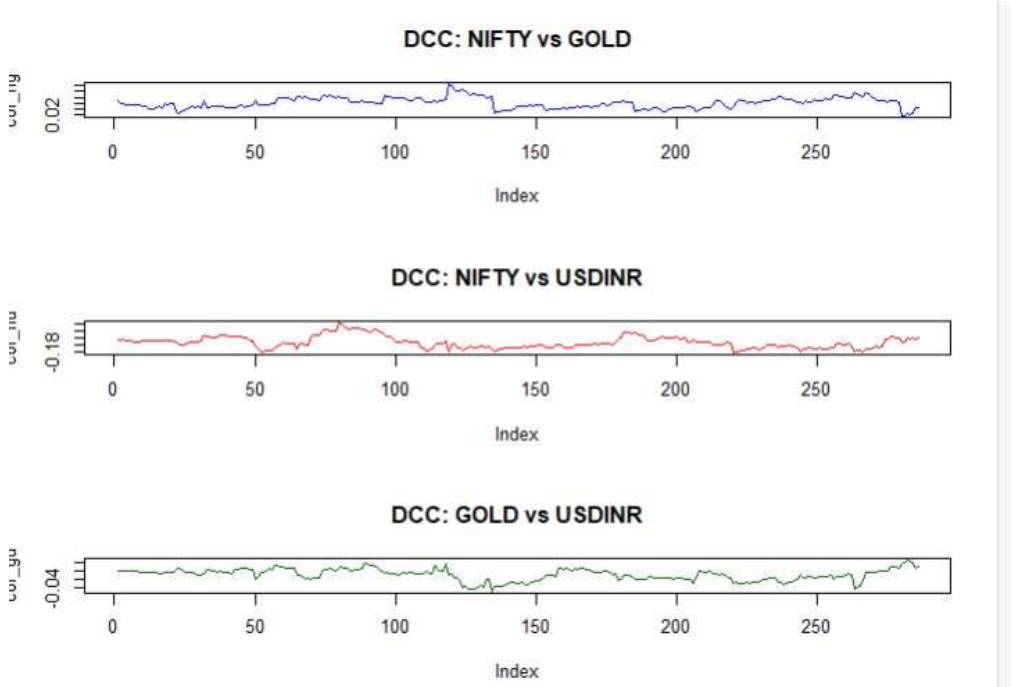


Figure 6: Dynamic Conditional Correlations — NIFTY vs Gold (blue), NIFTY vs USD/INR (red), Gold vs USD/INR (green). X-axis: observation index (October 2024 to January 2026).

The NIFTY-Gold DCC (blue series) traces a narrow band just above zero through most of the sample, consistent with the near-zero unconditional correlation established in Section 5.1 (Baur and McDermott, 2010). During the shock phase around April to June 2025, a discernible downward shift toward and briefly below zero is visible. This brief episode of negative NIFTY-Gold correlation is the empirical signature of the safe-haven rotation: as NIFTY fell sharply on April 2, 2025, investors simultaneously accumulated gold, generating a temporary inverse co-movement between equity and gold returns that is invisible to the static correlation measure (Chemkha et al., 2021; Hassan et al., 2022). The DCC subsequently recovers toward zero and stabilises through the recovery phase from July 2025 to January 2026, confirming that the safe-haven shift was temporary and reactive rather than a structural re-coupling of equity and gold markets.

The NIFTY-USD/INR DCC (red series) tells a structurally different story. This correlation is persistently and deeply negative throughout the entire sample, hovering around minus 0.18, reflecting the entrenched capital-flow channel: equity declines and rupee depreciation tend to co-occur as FPIs liquidate Indian equity positions and repatriate funds. During the shock phase,

a notable upward movement, meaning the correlation becomes less negative, is visible around the April to June 2025 window. This temporary weakening of the negative correlation likely reflects the RBI's active currency intervention dampening the rupee's response relative to the magnitude of the equity decline, partially decoupling the two series during the most acute phase of the shock (Chkili, 2012). By the recovery phase, the NIFTY-USD/INR DCC returns to its deeply negative pre-event level, confirming that the structural equity-currency relationship was not broken by the tariff shock, only temporarily disrupted.

The Gold-USD/INR DCC (green series) exhibits moderate negative values throughout the sample, consistent with the mechanical relationship between the rupee's exchange rate and the rupee price of internationally priced gold (Reboredo et al., 2016). During April to June 2025, the negative correlation intensifies as simultaneous rupee depreciation and gold price appreciation jointly drove this tightening, with both currency weakness and safe-haven demand elevating domestic gold prices during the uncertainty episode. From July 2025 onward, the Gold-USD/INR DCC trends progressively toward zero, reflecting the normalisation of both the exchange rate and gold prices as the shock was absorbed into the recovery phase.

Taken together, these DCC results make a critical contribution that no other method in this study can provide (Yadav et al., 2022). The absence of extreme or permanent spikes in any of the three dynamic correlations suggests that markets did not experience systemic contagion despite the shock. The inter-market linkages intensified, but did so in an orderly and bounded manner, consistent with rational portfolio adjustment rather than panic-driven contagion. Crucially, all three DCC series converged back toward their pre-shock baselines by the later portion of the sample, providing time-varying correlation evidence for the stabilisation predicted by Hypothesis H1.

5.7 Wavelet Power Spectrum Analysis

Figures 7, 8, and 9 present the individual Wavelet Power Spectra for NIFTY 50, Gold, and USD/INR respectively (Torrence and Compo, 1998). Time runs along the x-axis covering observations 1 to 286 from October 2024 to January 2026; period in trading days runs along the y-axis; colour encodes volatility power intensity; and black contour lines mark regions of statistical significance at five percent.

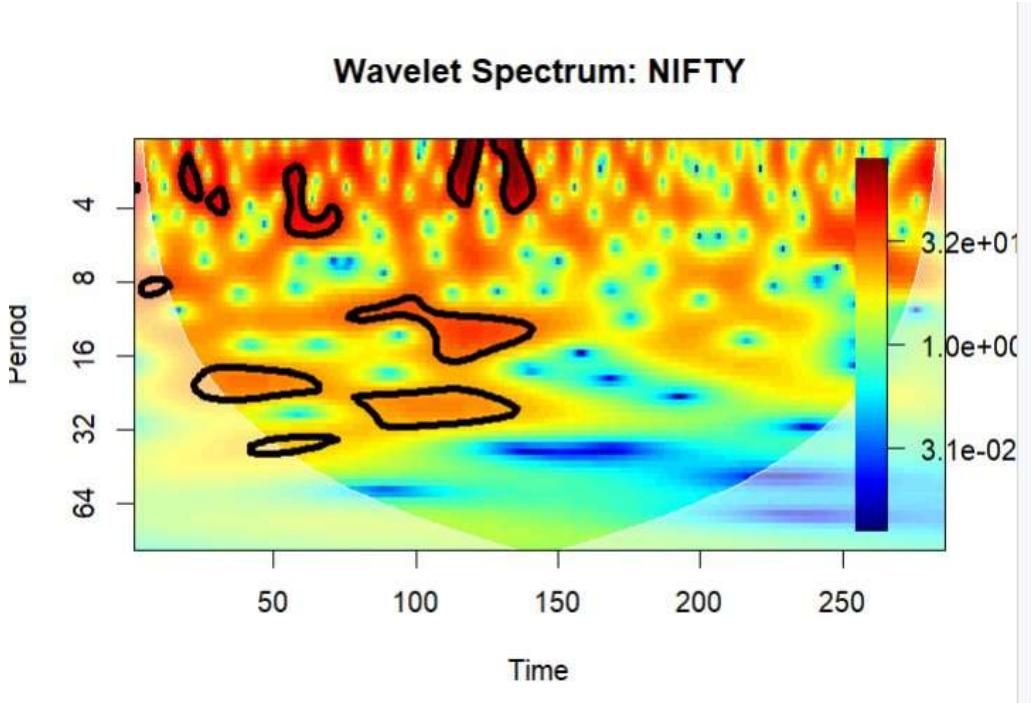


Figure 7: Wavelet Power Spectrum — NIFTY 50. Period (trading days) on y-axis. Red = high power, Blue = low power. Black contours = significance at 5%.

The NIFTY 50 spectrum (Figure 7) is well-behaved before the shock, showing scattered moderate-power activity at short periods consistent with routine equity market fluctuations. The shift at the onset of the shock phase is unmistakable. Significant volatility clusters are observed during April to June 2025, concentrated at medium-frequency periods of sixteen to thirty-two trading days, aligning directly with the tariff announcement and the weeks of institutional portfolio adjustment that followed (Sharma et al., 2022). This medium-frequency concentration is interpretively important: it suggests that the dominant timescale of the NIFTY response to the tariff shock was multi-week, reflecting the time it took for institutional investors including mutual funds, pension funds, and FPIs to reassess and rebalance Indian equity exposures. Through the recovery phase from July 2025 to January 2026, the NIFTY spectrum progressively cools, with no statistically significant contoured regions visible in the final quarter of the sample.

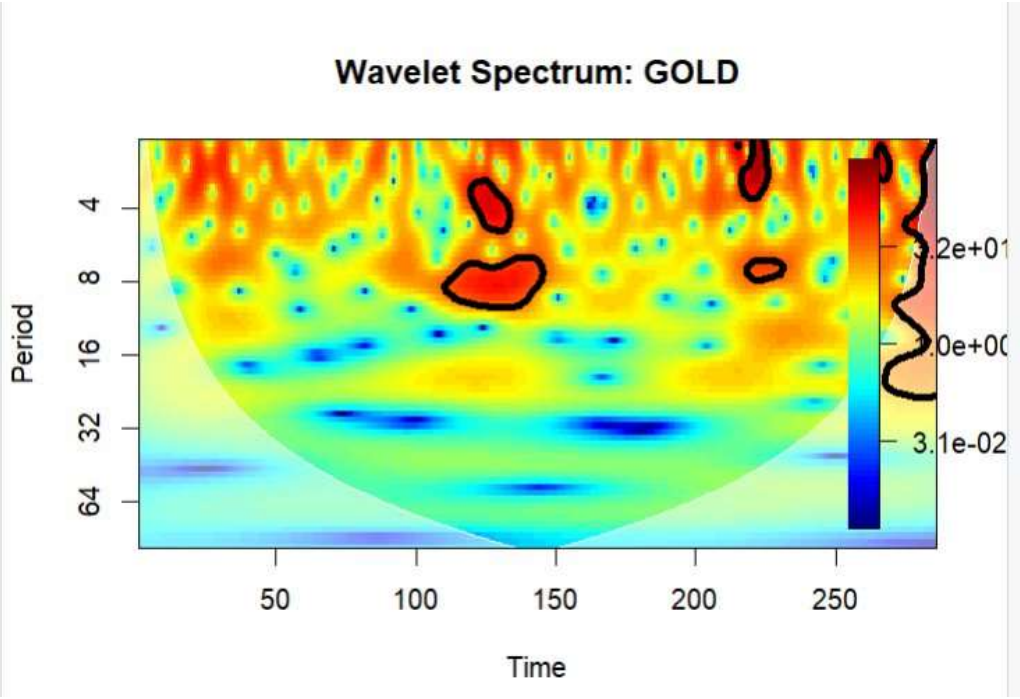


Figure 8: Wavelet Power Spectrum — Gold. Period (trading days) on y-axis. Red = high power, Blue = low power. Black contours = significance at 5%.

The Gold spectrum (Figure 8) presents a striking contrast (Aloui et al., 2022). Significant volatility clusters are observed during April to June 2025, but unlike NIFTY, these clusters are concentrated at short periods of four to eight trading days rather than medium-frequency horizons. This short-period power concentration is the frequency-domain signature of gold's reactive safe-haven role: investors rotated into gold rapidly, within days rather than weeks, following the April 2 announcement, generating intense but short-duration volatility bursts at the daily-to-weekly frequency scale (Baur and McDermott, 2010). A secondary region of significant power near the end of the sample likely reflects renewed global uncertainty. The gold spectrum's broader distribution across frequency scales compared to NIFTY reflects the asset's sensitivity to a wider and more heterogeneous set of global drivers, namely currency fluctuations, inflation expectations, central bank demand, and geopolitical risk, all operating simultaneously at different periodicities.

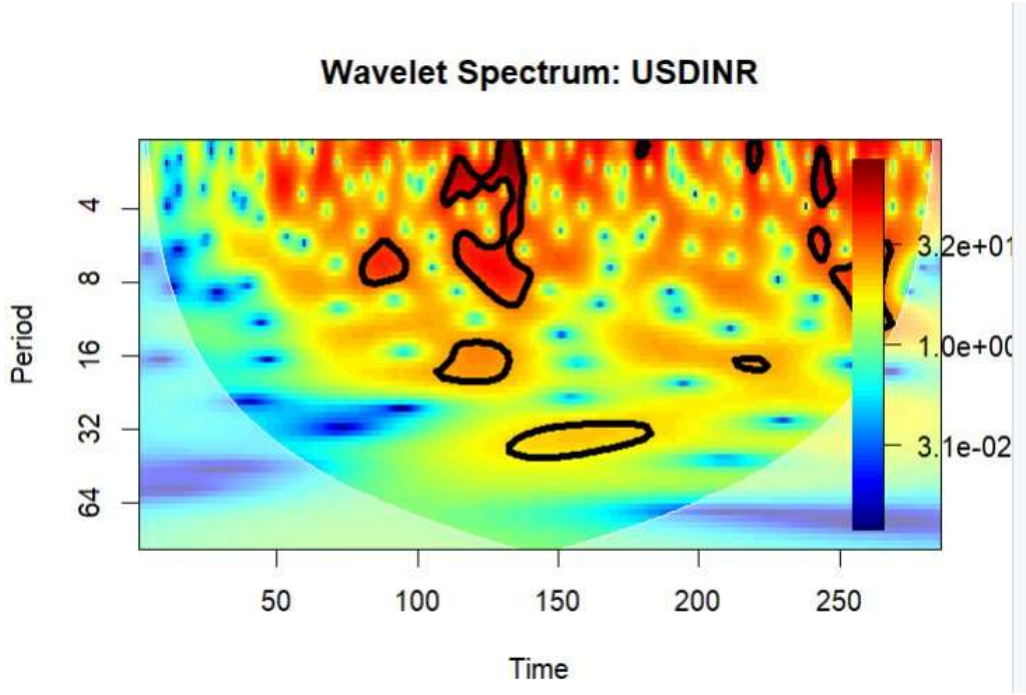


Figure 9: Wavelet Power Spectrum — USD/INR. Period (trading days) on y-axis. Red = high power, Blue = low power. Black contours = significance at 5%.

The USD/INR spectrum (Figure 9) is the most fragmented of the three and the most clearly shaped by institutional intervention (Chkili, 2012). Significant volatility clusters are observed during April to June 2025, concentrated at very short periods of two to four trading days, reflecting the sharp but brief currency depreciation pressure visible in Figure 3. The significant contoured region at the longer period of approximately thirty-two days during the same shock window captures the slower, sustained adjustment phase as currency traders and corporate hedgers recalibrated their longer-horizon positions. The episodic, fragmented character of the USD/INR spectrum even within the shock phase, with warm patches interrupted by cooler regions, directly reflects the RBI's active dampening of sustained volatility at any single frequency. As the recovery phase progresses from July 2025 onward, the USD/INR spectrum cools substantially, confirming the return of exchange rate volatility to its pre-shock managed baseline.

The comparative reading of the three spectra delivers the study's most market-specific finding (Rua and Nunes, 2009). Significant volatility clusters are observed during April to June 2025 across all three markets, directly aligning with the tariff announcement, but the frequency scales at which those clusters appear differ fundamentally. NIFTY's shock lives at medium frequencies, reflecting multi-week institutional processes. Gold's shock erupts at short frequencies, reflecting the rapid reactive dynamics of safe-haven demand. USD/INR's shock is

fragmented and attenuated at all frequencies by central bank management. These differences are entirely invisible to time-domain methods and constitute the unique informational contribution of the Wavelet Power Spectrum approach.

The findings of Section 5 collectively and strongly support Hypothesis H1. VAR analysis confirms that cross-market shock transmission operated simultaneously on April 2, 2025. IRF analysis confirms that mean-level impacts faded within three to five trading days, consistent with efficient price discovery (Fama, 1970). GARCH estimation confirms that volatility persistence is near-integrated across all three markets, with risk-level normalisation only occurring well into the recovery phase, a pattern consistent with comparable emerging market evidence (Sharma et al., 2022). DCC analysis confirms that cross-market correlations shifted meaningfully during April to June 2025 but converged back toward pre-shock baselines by the recovery phase, with no evidence of systemic contagion. Wavelet Power Spectrum analysis confirms that significant volatility clusters observed during April to June 2025 progressively dissipate as the recovery phase advances. The hypothesis that markets react immediately but stabilise over time is thus supported across all seven dimensions of the analysis.

6. **Conclusion**

This study set out to understand what happens to Indian financial markets when a major external trade shock arrives, not just on the day of the shock, but in the weeks and months that follow. The U.S. tariff announcement of April 2, 2025 provided the event; 286 daily observations across NIFTY 50, Gold, and USD/INR provided the data; and seven econometric methods provided the analytical lens. The answer that emerged is both clear and nuanced.

On April 2, 2025, all three markets moved simultaneously and in the directions economic theory would predict: equities fell, gold rose, and the rupee came under depreciation pressure. The VAR model confirms the simultaneity; the IRF analysis confirms that the mean-return impact of this shock was absorbed within roughly three to five trading days, consistent with the rapid price adjustment predicted by the Efficient Market Hypothesis (Fama, 1970). But GARCH estimation reveals a deeply different picture of risk: volatility persistence parameters approaching unity in all three markets mean that the elevated uncertainty environment created in April to June 2025 did not dissipate quickly. For NIFTY, meaningful risk normalisation came only months into the recovery phase. For Gold and USD/INR, the near-integrated GARCH processes imply that volatility shocks were essentially permanent within the sample window. This pattern aligns closely with evidence from emerging market GARCH studies,

where both standard and asymmetric specifications document similarly slow volatility decay (Sharma et al., 2022).

The DCC analysis adds a dimension that no other method could provide (Engle, 2002). Cross-market correlations shifted meaningfully during the shock phase: the equity-gold relationship turned temporarily negative as safe-haven rotation intensified, the equity-currency negative relationship briefly weakened as RBI intervention partially decoupled the two series, and the gold-currency correlation tightened as both depreciation and safe-haven demand jointly elevated domestic gold prices. But none of these shifts proved permanent. All three DCC series converged back toward their pre-shock baseline levels during the recovery phase from July 2025 onward, confirming that the Indian financial system's correlation architecture is resilient, even if its volatility levels are not quickly restored. The Wavelet Power Spectrum analysis completes the picture, revealing that the frequency fingerprint of the shock differs fundamentally across asset classes, with equity absorbing the disruption at medium horizons, gold responding at short horizons, and currency exhibiting a fragmented, institutionally managed pattern across all frequencies (Rua and Nunes, 2009; Aloui et al., 2022).

The findings strongly support the hypothesis that markets react immediately to trade policy shocks but stabilise over time as investors absorb information (Fama, 1970; Caldara et al., 2020). Price stability arrives quickly; volatility stability takes months; correlation stability lies in between. The central practical implication is straightforward: do not mistake the normalisation of prices for the normalisation of risk. Markets stabilise over time despite persistent volatility, and the tools to distinguish between the two, as demonstrated in this study, are available and informative.

7. **Managerial Implications**

The results of this study are not merely academic. They carry direct and actionable implications for the practitioners who navigate Indian financial markets.

For portfolio managers, the DCC findings are the most immediately useful (Yadav et al., 2022). The temporary improvement in gold's hedging effectiveness, specifically the brief episode of negative NIFTY-Gold correlation during April to June 2025, validates the tactical case for increasing gold allocation when trade policy uncertainty spikes (Baur and McDermott, 2010). But the subsequent return of the DCC to near-zero underscores an equally important point: that allocation should be unwound as correlations normalise, rather than maintained as a permanent structural shift. Managers who hold elevated gold weights well into the recovery phase are maintaining a hedge that has stopped working.

For risk managers, the near-integrated GARCH parameters should fundamentally reshape how stress testing and VaR calculations are conducted following external shocks (Sharma et al., 2022). Models that assume volatility mean-reverts within days will systematically understate tail risk for the months following a major tariff event. The NIFTY volatility half-life of approximately 55 trading days provides a concrete benchmark: risk budgets, margin requirements, and capital adequacy frameworks should remain elevated for at least this long after a comparable shock.

For foreign portfolio investors, the persistent deeply negative NIFTY-USD/INR DCC, even in its temporarily weakened shock-phase form, confirms the structural importance of currency hedging for Indian equity exposures (Chkili, 2012). The brief weakening of the equity-currency relationship during the shock phase, rather than being reassuring, signals the danger of relying on static hedge ratios that assume a constant correlation structure.

For regulators and policymakers at SEBI and the RBI, the study's findings validate existing cross-market surveillance frameworks while pointing to their limitations. The absence of systemic contagion, confirmed by the bounded, orderly character of the DCC dynamics, is reassuring. But the prolonged individual-market volatility documented by GARCH (Sharma et al., 2022), combined with the frequency-specific shock signatures revealed by Wavelet analysis, suggests that regulatory monitoring should extend beyond aggregate volatility measures to include time-varying correlation tracking and frequency-domain early warning indicators.

8. **Limitations and Future Research**

8.1 Limitations

This study should be read with several limitations in mind. The time horizon of approximately fifteen months is relatively short by the standards of financial econometrics, and the GARCH and DCC persistence parameters estimated on such a window may be sensitive to the specific shock event that dominates the sample. A longer pre-event baseline would have provided more stable unconditional benchmarks. The reliance on daily frequency data means that the within-day dynamics on April 2, 2025 itself, specifically the minutes and hours immediately following the tariff announcement, are entirely invisible to the methodology. The focus on a single event limits the generalisability of the findings to other types of trade policy shocks; quotas, export controls, and bilateral negotiations may generate quite different market dynamics than a broad tariff announcement. The use of aggregate market indices rather than sector-level or firm-level data masks the heterogeneous impact across industries with different tariff exposures. The IT

sector, the pharmaceutical sector, and the automobile sector each face distinct vulnerabilities that the NIFTY 50 composite cannot reveal. Finally, the standard DCC specification assumes a constant unconditional correlation matrix, which may not fully capture structural breaks in the inter-market relationship architecture (Engle, 2002). Additionally, while this study employs the standard sGARCH(1,1) specification, the use of non-linear alternatives such as EGARCH or GJR-GARCH could provide further insight into asymmetric volatility dynamics, particularly for USD/INR, where mild sign bias was detected (Sharma et al., 2022).

8.2 Future Research

Each limitation points toward a productive research direction. Sector-level DCC and Wavelet analysis comparing NIFTY IT, Pharma, and Auto would reveal heterogeneous dynamics invisible at the index level. Asymmetric GARCH specifications, specifically EGARCH or GJR-GARCH, would address the sign-bias evidence for USD/INR and the potential asymmetry in gold's response to positive versus negative shocks (Sharma et al., 2022). The Go-GARCH or A-DCC model would relax the constant unconditional correlation assumption, allowing for more flexible structural break accommodation in the correlation architecture. High-frequency intraday data around April 2, 2025 would enable within-day price discovery analysis, capturing the minutes immediately following the tariff announcement that lie outside the reach of daily data (Fama, 1970). A multi-country comparative study applying the same DCC and Wavelet Power Spectrum framework to other April 2025 tariff-affected emerging economies such as Vietnam, Bangladesh, and Mexico would determine whether India's pattern of bounded, transient contagion is idiosyncratic or generalisable.

References

Aloui, R., Hammoudeh, S., & Nguyen, D. K. (2022). A time-frequency analysis of co-movements across commodity and equity markets. Journal of International Financial Markets, Institutions and Money, 80, 101646.

Alfaro, L., Chari, A., Greenland, A., & Schott, P. (2020). Aggregate and firm-level stock returns during pandemics in real time. NBER Working Paper No. 26950.

Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. Quarterly Journal of Economics, 131(4), 1593–1636.

Baur, D. G., & McDermott, T. K. (2010). Is gold a safe haven? International evidence. Journal of Banking and Finance, 34(8), 1886–1898.

Bekaert, G., Ehrmann, M., Fratzscher, M., & Mehl, A. (2014). The global crisis and equity market contagion. Journal of Finance, 69(6), 2597–2649.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. Journal of Econometrics, 31(3), 307–327.

Caldara, D., Iacoviello, M., Molligo, P., Prestipino, A., & Raffo, A. (2020). The economic effects of trade policy uncertainty. Journal of Monetary Economics, 109, 38–59.

Chakraborty, I., & Sen, B. (2020). Volatility patterns in Indian stock markets post policy announcements: A GARCH model analysis. Indian Journal of Economics and Development, 16(2), 132–142.

Chemkha, R., BenSaida, A., Ghorbel, A., & Tayachi, T. (2021). Hedge and safe haven properties during COVID-19: Evidence from Bitcoin and gold. Quarterly Review of Economics and Finance, 82, 71–85.

Chkili, W. (2012). The dynamic relationship between exchange rates and stock returns in emerging countries: Volatility spillover and portfolio management. International Journal of Management Science and Engineering Management, 7, 253–262.

Engle, R. F. (1982). Autoregressive conditional heteroskedasticity with estimates of the variance of U.K. inflation. Econometrica, 50(4), 987–1007.

Engle, R. F. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. Journal of Business and Economic Statistics, 20(3), 339–350.

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. Journal of Finance, 25(2), 383–417.

Goodell, J. W., Kumar, S., Lim, W. M., & Pattnaik, D. (2022). Artificial intelligence and machine learning in finance. Journal of Behavioral and Experimental Finance, 32, 100577.

Hassan, M. K., Kamran, M., Djajadikerta, H. G., & Choudhury, T. (2022). Search for safe havens and resilience to global financial volatility. Pacific-Basin Finance Journal, 73, 101768.

Li, Z., & Meng, Q. (2022). Time and frequency connectedness and portfolio diversification between cryptocurrencies and renewable energy stock markets during COVID-19. North American Journal of Economics and Finance, 59, 101565.

Markowitz, H. (1952). Portfolio selection. Journal of Finance, 7(1), 77–91.

Narayan, P. K., Rizvi, S. A. R., & Sakti, A. (2022). Did green debt instruments aid diversification during the COVID-19 pandemic? Financial Innovation, 8(1), 1–15.

Rao, A., Bhardwaj, S., & Patel, R. (2024). Spillover effects of U.S. policy shocks on Indian markets. Emerging Markets Review, 50, 100890.

Reboredo, J. C., Rivera-Castro, M. A., & Ugolini, A. (2016). Downside and upside risk spillovers between exchange rates and stock prices. Journal of Banking and Finance, 62, 76–96.

Rua, A., & Nunes, L. C. (2009). International comovement of stock market returns: A wavelet analysis. Journal of Empirical Finance, 16(4), 632–639.

Sharma, S., Aggarwal, V., & Yadav, M. P. (2022). Comparison of linear and non-linear GARCH models for forecasting volatility of select emerging countries. Journal of Advances in Management Research, 19(4), 526–547.

Sims, C. A. (1980). Macroeconomics and reality. Econometrica, 48(1), 1–48.

Singh, R., Mehta, P., & Suri, D. (2023). Foreign portfolio investment and volatility transmission: Evidence from India during U.S. tariff uncertainty. International Review of Economics and Finance, 87, 304–317.

Torrence, C., & Compo, G. P. (1998). A practical guide to wavelet analysis. Bulletin of the American Meteorological Society, 79(1), 61–78.

Yadav, M. P., Sharma, S., & Bhardwaj, I. (2022). Volatility spillover between Chinese stock market and selected emerging economies: A dynamic conditional correlation and portfolio optimization perspective. Asia-Pacific Financial Markets, 29, 1–18.

National Stock Exchange of India. (2025). Historical index data. Retrieved from https://www.nseindia.com

Office of the United States Trade Representative. (2025). Tariff policy press releases. Retrieved from https://ustr.gov

Yahoo Finance. (2025). Historical market data: NIFTY 50, Gold Futures, USD/INR. Retrieved from https://finance.yahoo.com